

Exploring the biogeographical diversity of *Trichoderma* for plant health

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24.1 Introduction

A thorough knowledge of the systematic of *Trichoderma* is highly imperative to discriminate between the beneficial and pathogenic species of *Trichoderma* and to protect the diversity (Abdelrahman et al., 2016). Hence, in-depth knowledge of *Trichoderma* on the present existence of the diversity, where do they exist? and any threat to them has to be well documented to explore their potential for human welfare (Nagaraju et al., 2012b). Though morphological taxonomy helps in the differentiation of strains, the era of omics adds value to the phenotypic description of the strains and helps in documenting and differentiating the diversity. Hence, the systematic of *Trichoderma* should be carried in combination with morphological, biochemical, and omic approaches in a polyphasic manner. Biodiversity studies on *Trichoderma* conducted in European countries, Russia, and China reflect the presence of 62 teleomorphs of *Trichoderma* (Hypocrea) and 25 species of *Trichoderma* (Anamorphic of Hypocrea). The biodiversity of *Trichoderma* from different habitats plays a crucial role in the management of plant diseases, bioremediation and thus improves human welfare.

24.2 Is *Trichoderma* important?

The need for the agricultural community was the force of motivation for the development of broad-spectrum pesticides to manage pests and diseases of both agricultural and horticultural crops (Jogaiah et al., 2013). Advances in the area of pesticide development were measured by the assessment of its efficacy in the management alone. However, today the food production has ceased to be a major concern, due to stable population levels and the success of innovative agricultural technologies. At this juncture, the agrochemical industry that primarily recognized the farmers as their customers have failed to acknowledge the public as an important client. The perception of the public reflects that the negative aspects of pesticides seem to

outnumber their benefits. The increased reflection on environmental concern over pesticide use has been instrumental in a large upsurge of bio-agents for both pest and disease control (Jogaiah et al., 2018). The development of fungicide resistance among the pathogens, groundwater, and foodstuff pollution and the development of oncogenic risks have further encouraged the exploitation of antagonistic microflora in disease management (Satapute et al., 2019).

The decades of laboratory experiments have led to the explosion in the field of bio-control and introduced more than 50 commercial formulations comprising of fungi and bacteria (Whipps, 1997; Murali et al., 2013). Among the antagonists, *Trichoderma* plays a vital role in plant disease management and has opened new vistas for the commercialization. *Trichoderma* spp. are cosmopolitan soil fungi, remarkable for their rapid growth, capable of utilizing diverse substrates, and resistance to noxious chemicals (Nagaraju et al., 2012b). It is a predominant component among the mycoflora in various soils in all climatic zones. The frequent association of *Trichoderma* (*Hypocreales*, *Ascomycota*) with decaying wood produces more enzymes of industrial value (*Trichoderma reesei*=*Hypocrea jecorina*) (Kubicek and Penttilä, 1998; Sudisha and Shekar Shetty, 2009) and antibiotics (Sivasithamparam and Ghisalberti, 1998) or has applied as biocontrol agents against plant pathogens (i.e., *T. harzianum*=*H. lixii*; *T. atroviride*=*H. atroviridis*; *T. asperellum*, *T. virens*) (Hjeljord and Tronsmo, 1998; Jogaiah et al., 2018). More recently, *T. longibrachiatum* has also known as an opportunistic pathogen of immunocompromised mammals including humans (Kredics et al., 2003). *Trichoderma* has been recognized to comprise a significant amount of fungal biomass in soil (Nelson, 1982) and is frequently present as an indoor contaminant (Thrane et al., 2001). These diverse implications of *Trichoderma*/*Hypocrea* with human society render an accurate species and strain identification as paramount importance (Abdelrahman and Jogaiah, 2019). However, classical approaches based on the use of morphological criteria are difficult to apply to *Trichoderma* due to the plasticity of characters. Hence, the knowledge on the diversity and distribution pattern of *Trichoderma* is highly imperative to explore the best among the diverse species of *Trichoderma*. The analysis of the *Trichoderma* products registered with CIB, Faridabad, and Haryana revealed that only *T. viride*, *T. harzianum*, and *T. hamatum* have been explored to a maximum extent.

24.3 Attributes of *Trichoderma* as a successful biocontrol organism

Trichoderma is the most widely used biocontrol agent. Among the various isolates of *Trichoderma*, *T. viride*, *T. harzianum*, *T. virens*, and *T. hamatum* are used for the management of various diseases of crop plants (Nagaraju et al., 2012a,b; Jogaiah et al., 2013).

It has many attributes to call it as a best suited biocontrol agent, which are:

- a. High rhizosphere competence
- b. High-competitive saprophytic ability

- c. Enhance plant growth
- d. Great arsenal of inducible polysaccharide—degrading enzymes
- e. Ease for mass multiplication
- f. Broad spectrum of action against various pathogens
- g. Provide reliable control
- h. Environmentally safe

24.4 Ecology of *Trichoderma*

The significance of genetic variation is a key factor to evaluate biodiversity. The genetic variations in *Trichoderma* are mainly attributed to the adaptation of different species suited to different ecological conditions. The hymenomycetous genus, *Trichoderma* is a predominant cosmopolitan soil fungus, widely distributed all over the world, well known for its rapid growth and has the potential to use the varied substrates and often they are resistant to toxic chemicals. *Trichoderma* occurs in all soils and other natural habitats that are rich in organic matter. *Trichoderma* is also found on root surfaces of various plants, decayed bark, mainly after the decay by other fungi (Esposito and da Silva, 1998). The population of *Trichoderma* decreases when it is exposed to dry conditions for a long period of time. Some strains of *T. hamatum* and *T. pseudokoningii* are adapted to excessive soil moisture and those of *T. viride* and *T. polysporum* are not found to be restricted to areas of low temperature. *T. harzianum* is predominant in warm climatic regions, while *T. hamatum* and *T. koningii* are widely distributed in the areas of diverse climatic conditions (Papavizas, 1985; Grondona et al., 1997). It is also distributed in different soils including agricultural, forest, prairie, salt marsh, and desert soils distributed in different agroclimatic zones. The propagules of *Trichoderma* were abundant in forest soils and they proliferated well in the forest soils rich in carbon and nitrogen (Danielson and Davey, 1973). Some of the species of *Trichoderma* survive as decomposers of woody and herbaceous materials and are also necrotrophic against the primary wood decomposers (Rossman, 1996).

The population dynamics of *Trichoderma* are also altered due to the variations in the iron content of the soil (Xu, 1996). Certain strains of *Trichoderma* have evolved and have occupied the niches of higher elevations. Analysis of soil samples from higher elevations of about 2700m in the Himalayas reflected the presence of section *Longibrachiatum*. Based on the parsimony analysis the same was grouped under *H. audinensis*. The same was also isolated from 2300 m elevation of Venezuelan Andes (Samuels et al., 1998). Another member of the section *Longibrachiatum* namely, *T. konilangbra* was isolated from locations between 1750 and 3400m in Uganda.

24.4.1 Ecology-based diversity of *Trichoderma*

The diversity of the species of *Trichoderma* varies based on locations and with the fauna associated with the microsites. The soils of Taiwan were rich in *T. asperellum*, *T. sp. novum* was found in association with tree barks of Taiwan, and tree bark with

lichen in Chitan forest of Taiwan had *Hypocrea jecorina*, *T. viride*, *T. atroviride*, and *T. harzianum*, but the tree barks that were in association with moss were found to contain *T. harzianum* alone.

Similarly, the tropical forest soil of Cambodia was rich in *T. harzianum*, indicating that it prefers warm soil temperature for its survival and multiplication. Analyses of the samples from the ruined walls of the temple indicate that they were predominant with *T. harzianum* and to a certain extent the ruined walls were also found to be associated with *T. virens*. The soils near the seashore of Thailand at Coral Island and Koh Samet Island was colonized by *T. sp. novum* and *T. harzianum*. The wet soils of Kanchanaburi province of Burma were colonized by *T. harzianum* and *T. ghanense*. The tropical rain forest soils of Malaysia had *T. virens*, *T. spirale*, *T. harzianum*, *T. asperellum*, and *T. sp. novum*. The lake garden soils of Kuala Lumpur were associated with *T. virens*, *T. spirale*, *T. harzianum*, *T. sp. novum*, *T. koningii*, and *T. atroviride*. The forest soil of Singapore was colonized by *T. koningii*, *T. asperellum*, *T. virens*, *T. spirale*, and *T. harzianum*. Though there was a huge variation in the distribution pattern of the *Trichoderma* spp., the cultivated garden soils of South East Asia were predominant with *T. virens* which was used as biocontrol fungi for the management of soilborne diseases. This vividly explains that the ecological requirements are different from the species colonizing tree bark, forest soils, and cultivated garden soils (Kubicek et al., 2003).

Isolates of the *T. spirale* was found to be more in South East Asia than the cosmopolitan species *T. koningii*. Similarly, several isolates of *T. spirale* were observed in the soils of Canada. However, the abundance of the same was more in Central America, Columbia, and South Africa which suggests that this species is more abundant in tropical climates (Kubicek et al., 2003). *H. jecorina* a pantropical ascomycete is abundant in South-East Asia, The Pacific Isles, and South America, but the same was previously reported to be restricted to the regions around equator (Samuels et al., 1998). Soils from Central Russia, the Urals, Siberia, and the Himalayan mountains were predominant with *T. atroviride*, *T. hamatum*, *T. virens*, *T. asperellum*, *T. koningii*, *T. harzianum*, and *T. oblongisporum* (Kullnig et al., 2000).

24.5 Systematics of *Trichoderma* and its significance in biodiversity

Knowledge of *Trichoderma* systematics is highly essential to explore the biodiversity of *Trichoderma*. *Trichoderma* is a complex group of fungi that are used in the production of cellulolytic and hemicellulolytic enzymes, biocontrol of plant diseases, and biodegradation of chlorophenolic compounds and soil bioremediation. Since *Trichoderma* is being used as a very common beneficial fungus for several applications (Nandini et al., 2020), and the systematics of *Trichoderma* becomes very imperative in the identification and utilization of various strains. For this purpose, several molecular and morphological approaches are available for the identification of various isolates.

Trichoderma was introduced by Persoon (1794). The taxonomy and identification of *Trichoderma* remain inconclusive. All strains of *Trichoderma* were identified as “*T. viride*” till 1969 (including all the cellulase-producing strains of *T. reesei*) owing to Bisby’s (1939) concept that *Trichoderma* consists of a single species. This led to the erroneous statement even in textbooks that “*T. viride* is an industrial cellulose producer.” Hence, most of the taxa sampled and determined before 1969 were misidentified. Rifai (1969) adopted the concept of ‘aggregate’ species, distinguished nine ‘species aggregates,’ and admitted that some of them (particularly *T. hamatum*) likely contain two or more morphologically indistinguishable species. Bissett (1984, 1991a,b,c, 1992) and Gams and Bissett (1998) revised Rifai’s aggregate species by recognizing the continuities of morphological characters, expanded the morphological criteria to accommodate the wider range of morphological variation expressed by some anamorphs of *Hypocrea*, and adopted some forms previously included in *Gliocladium*. While dissecting several of Rifai’s species aggregates into several defined taxa, Bissett (1991a) established a subdivision of the genus into five sections: *Longibrachiatum*, *Pachybasium*, *Trichoderma*, *Saturnisporum*, and *Hypocreanum*. The advent of molecular tools for investigations in fungal taxonomy prompted research in the mid-1990s to reassess the morphology-based taxonomy in *Trichoderma*. The laboratories of G.J. Samuels (Beltsville, MD, USA), T. Börner (Berlin, FRG), and C.P. Kubicek (Vienna, Austria) collaboratively pioneered a revision of Bissett’s section *Longibrachiatum*. They combined the use of molecular markers (ITS1 and ITS2 sequence analysis, RAPD), physiological (isoenzyme analysis), and phenetic characters, and also for the first time included an analysis of potential teleomorphs of the *Trichoderma* spp. from this section (Samuels et al., 1998). As a result, section *Longibrachiatum* was recognized to be monophyletic and to contain 10 taxa, within which four pairs displayed teleomorph-anamorph relationships: *H. schweinitzii*/T. citrinoviride; *H. pseudokoningii*/T. pseudokoningii; *H. jecorinal*/T. reesei; and *H. orientalis*/T. longibrachiatum.

Further merged section *Saturnisporum* included only two species, *T. saturnisporum* and *T. ghanense* (Doi et al., 1987), with section *Longibrachiatum*, and recognized the synonymy of *T. ghanense* with *T. parceramosum*. Yet, as a whole, the concept of section *Longibrachiatum* as defined by Bissett (1984) was largely confirmed, suggesting a degree of correlation between morphological and molecular approaches to taxonomy in *Trichoderma*. Section *Longibrachiatum* is a comparably small section of *Trichoderma*, and phylogenetically most distant from the other sections. The relationship between morphological characters and molecular phylogeny became more complex, however, when larger sections of *Trichoderma* were investigated. Kindermann et al. (1998) attempted a first phylogenetic analysis of the whole genus. Using sequence analysis of the ITS1 region of rDNA, they found that the largest section, *Pachybasium*, is actually paraphyletic. Although the use of a single gene fragment alone is insufficient by today’s standards, this finding has been confirmed by phylogenetic analysis of several other genes (Kullnig-Gradinger et al., 2002; Chaverri et al., 2004). In this context the nomenclatural-type strain of sect. *Pachybasium*, *Pachybasium hamatum* (Bonord.) = *T. hamatum*, is not a member of the major one of the two *Pachybasium* clades (clade B) (Kindermann et al., 1998), but clusters

together with *T. pubescens* and *T. strigosum* in a clade otherwise containing almost all taxa (i.e., with the exception of *T. aureoviride*) from section *Trichoderma*. Because of the lack of a distinctive morphological hiatus between clades A and B of *Pachybasium* (Kindermann et al., 1998), researchers have so far maintained the name *Pachybasium* for both clades, but it is clear that this is an unsatisfactory situation from a taxonomic point of view. *Pachybasium* B contains all taxa attributed to this section by Bissett (1991b) with the exception of the three mentioned above. In addition, *Pachybasium* B also poses the problem of strong genetic variability of several of its species (e.g., *T. harzianum*). Since its species (both ana- and teleomorphs) account for the majority of taxa found in field investigations (Gherbawy et al., 2004), this raises the question of how to identify species in this large heterogeneous group.

24.5.1 Morphological taxonomy of *Trichoderma*

Morphological taxonomy was based on colony morphology, characters of conidiophore, phialides, phialospores, and chlamydospores.

24.5.1.1 Key to *Trichoderma* genus

Colony

- Colonies rapidly growing
- Mycelium mostly submerged
- Mycelium may be matte or floccose or woolly or arachnoid aerial mycelium
- The reverse side of the plate may be buff or yellowish or amber or reddish or yellow-green or colorless
- Conidiation effuse/tufted/compact pustules; white at first, remaining as such or turns green, gray, or brown

Chlamydospores

- Abundant in submerged mycelium; intercalary or terminal on short lateral branches of vegetative hyphae
- Globose to ellipsoid, smooth or thick walled, colorless to pale yellowish or greenish

Conidiophores

- The main axis is broad, straight, or flexuous
- Primary branches arise at regular intervals and produce smaller secondary branches of branching nature
- Branches are more or less divergent, solitary, paired, or in verticils
- Repeated verticillate branching results in the pyramidal structure
- In some species, the main axis of the conidiophore ends in sterile structures which might be simple, branched, or coiled

Phialides

- Flask shaped, slightly narrower at the base, bulged at the middle and with narrow conical or subcylindrical neck
- Phialides are disposed of in verticils or whorls beneath the septa of the conidiophore

Phialospores

- Conidia single celled, colorless/grayish/green or brown
- Cell wall may be smooth/roughened or with wing-like projection on the outer wall
- Conidial shape may be subglobose, obovoid, ellipsoid, oblong, or short cylindrical [Bissett's \(1991a,b,c\)](#)

24.5.2 Sections of *Trichoderma*

The genus *Trichoderma* is divided into five sections. They are

- *Trichoderma*
- *Pachybasium*
- *Longibrachiatum*
- *Saturnisporum*
- *Hypocreanum*

24.5.2.1 *Trichoderma*

- The species in section *Trichoderma* have narrow and flexuous conidiophores
- Branches of conidiophore and phialides are uncrowded
- Branches are frequently paired and never in verticils of more than three.
- (*T. koningii*, *T. aureoviride*, *T. viride*, *T. atroviride*)

24.5.2.2 *Pachybasium*

- Highly ramified colonies
- Conidiophores are broad, arranged in compact pustules, or fascicles
- Branches and phialides are broad or inflated, relatively short
- Phialides on the branches are distributed in crowded verticils
- Few species have the extension of sterile conidiophore
- Many of them produce compact conidiogenous pustules by anatomizing with adjacent conidiophores ([Rifai, 1969](#))
(*T. hamatum*, *T. harzianum*, *T. piluliferum*, *T. polysporum*)

24.5.2.3 *Longibrachiatum* ([Bissett, 1984](#))

- Distinct greenish-yellow pigments are produced on the reverse of the cultures
(*T. pseudokoningii*, *T. longibrachiatum*)

24.5.2.4 *Saturnisporum*

- Branching system is characterized by *T. viride*
- Uncrowded branches and phialides
- Frequently paired branches
- Compact conidiogenous pustules
- Wing like or bullate conidial ornamentation
(*T. saturnisporum*)

24.5.2.5 *Hypocreanum* (teleomorph of *Trichoderma*)

Isolates from soil and wood pertain to this section

- Conidiation is effuse and sparse
- Conidiophores are sparingly branched
- Phialides are cylindrical to subulate
- They are borne in divergent verticils like verticillium [Bissett's \(1991a,b,c\)](#)

24.6 Global diversity of *Trichoderma*—An overview

Trichoderma being cosmopolitan fungi found to be widely distributed in different agroclimatic zones and habitats in both tropical and subtropical countries. However, the biogeographical distribution of *Trichoderma* is not concentrated except in European countries, Russia, and China. Since *Trichoderma* has a wide scope and potential to improve human life from different angles, attention must be focused to identify the diversity and distribution pattern of *Trichoderma*. About 35 *Trichoderma* species are recognized on the basis of morphological and molecular characters. Besides the role of a few of these species in biotechnology, several seem to play prominent roles in soil ecosystems. The diversity of *Trichoderma* spp. in Russia, Siberia, and the Himalayan Mountains were identified through molecular approaches using ITS 1 and 2 sequence of the rDNA cluster of the 75 isolates obtained. The comparison of sequences confirmed the presence of *T. atroviride*, *T. virens*, *T. hamatum*, *T. asperellum*, *T. koningii*, and *T. oblongisporum*, *T. harzianum*, and *T. inhamatum*. Section *Longibrachiatum* and *T. stromaticum* were found distributed at an elevation of 2700m in the Himalayas ([Kullnig et al., 2000](#)).

Parsimony and distance analyses of DNA sequences from multiple genetic loci of *Trichoderma* with 18S rDNA sequence analysis suggested that the genus *Trichoderma* might have evolved 110 million years ago as *Hypomyces* and *Fusarium*. 28S rDNA sequence analysis shows that the genus *Trichoderma* is part of a monophyletic branch within the *Hypocreaceae*. Gene trees inferred from a combined analysis of the nuclear ribosomal internal transcribed spacer provided strong statistical support for a phylogeny consistent with the existence of four clades. Clade A comprises species of Bissett's (1991) sect. *Trichoderma* in addition to *T. hamatum*, *T. pubescens*, *T. asperellum*, and *T. strigosum*; clade C comprises all the species contained in section *Longibrachiatum* as revised by [Samuels et al. \(1998\)](#), and clade D contains only *T. aureoviride* which is genetically most distant to all other species. Clade B, on the other hand, contains a large and taxonomically heterogeneous mixture of species, among which several subclades could be identified: subclade B1 containing *H. lactea*, *H. citrina*, *H. citrina* var. *americana*, *H. lutea*, and an unnamed *T. sp. 1*; subclade B2 containing *T. stromaticum*, and an unnamed *T. sp. PPRI3559*; subclade B3 containing *T. fertile*, *T. oblongisporum*, and *H. hunua*; subclade B5 containing *T. polysporum*, *T. roceum*, *Hypocrea pilulifera*, and *T. minutisporum*; and a larger subclade (B4), in which three strain clusters could be distinguished: one comprising *T. harzianum*, *T. inhamatum*, *H. vinosa*, and *T. aggressivum*, another one containing *T. fasciculatum*,

T. longipile, and *T. strictipile*; and a third containing *T. virens*, *T. avofuscum*, and *T. crassum*. The position of the remaining species of subclade B4 (*T. spirale*, *T. cf. aureoviride*, *H. tawa*, and *T. tomentosum*) was not resolved (Kullnig-Gradinger et al., 2002).

The variation in *Trichoderma* in South East Asia (Taiwan and Indonesia) was identified through morphological, biochemical, and through sequence analysis. A total of 96 strains were isolated. In all, 78 isolates were identified as *Trichoderma harzianum*/*Trichoderma inhamatum* (37 strains) *Trichoderma virens* (16 strains), *Trichoderma spirale* (8 strains), *Trichoderma koningii* (3 strains), *Trichoderma atroviride* (3 strains), *Trichoderma asperellum* (4 strains), *Hypocrea jecorina* (anamorph: *Trichoderma reesei*; 2 strains), *Trichoderma viride* (2 strains), *Trichoderma hamatum* (1 strain), and *Trichoderma ghanense* (1 strain). Analysis of biochemical characters revealed that *T. virens*, *T. spirale*, *T. asperellum*, *T. koningii*, *H. jecorina*, and *T. ghanense* formed well-defined clusters, thus exhibiting species-specific metabolic properties. In addition, two new taxa of *Trichoderma* were identified (Kubicek et al., 2003).

The biodiversity and biogeography of *Trichoderma* in four areas of China: North (Hebei province), South-East (Zhejiang province), West (Himalayan, Tibet), and South-West (Yunnan province) include 135 isolates. They were grouped tentatively based on morphological characters which showed the existence of 64 isolates. But, all were grouped into 11 species as *Trichoderma asperellum*, *T. koningii*, *T. atroviride*, *T. viride*, *T. velutinum*, *T. cerinum*, *T. virens*, *T. harzianum*, *T. sinensis*, *T. citrinoviride*, *T. longibrachiatum*, and two putative new species (*T. sp. C1*, and *T. sp. C2*) by both morphological characters and phylogenetic analysis. *T. harzianum* accounted for almost half of the biodiversity and was predominant in North China. In addition, a unique haplotype of *T. harzianum* which is rarely found elsewhere was identified (Zhang et al., 2005).

24.7 Species diversity of *Trichoderma*

Trichoderma being cosmopolitan fungi, they are often the predominant components of the soil mycoflora in various soils, such as agricultural, prairie, forest, salt marsh, and desert soils in all climatic zones where they are involved in a number of processes like humic acid synthesis and degradation of xenobiotics (Klein and Eveleigh, 1998). Some are also used as good biocontrol agents to improve plant health (Joshi et al., 2019). There is a huge diversity of *Trichoderma* species in different soils and habitats. The diversity of *Trichoderma* from different geographical locations will be ascertained on the basis of morphology, physiological characteristics, and ITS1 and 2 sequences. Nucleotide variation in ITS has recently been used to identify most species and clades of *Trichoderma* that have been recognized by combined analyses of nuclear and mitochondrial rDNA plus fragments of two genes, translation elongation factor and endochitinase 42 (Kullnig-Gradinger et al., 2002). Various species of identified *Trichoderma* include:

Trichoderma viride, *T. harzianum*, *T. hamatum*, *T. cerinum*, *T. citrinoviride*, *T. asperellum*, *T. aggressivum* f. *aggressivum*, *T. cerinum.*, *T. atroviride*, *T. koningii*, *T. konilangbra*, *T. pseudokoningii*, *T. inhamatum*, *T. virens*, *T. velutinum.*, *T. spirale*, *T. reesei*, *T. ghanense*, *T. spirale*, *T. sinensis.*, *T. rossicum*, *T. longibrachyatum*, *T. pubescens*, *T. tomentosum.*, *T. strigosum*, *T. saturnisporum*, *T. erinaceum*, *T. effusum*, *T. atroviride*, *T. polysporum*, etc. The anamorphs of *Trichoderma* species are *Hypocrea aureoviridis*, *H. andinensis*, *H. jecorina*, *H. orientalis*, *H. novazelandia*, *H. tawa.*, *H. lixli.*, *H. cinnamomea.*, *H. catoptron.*, *H. thailandica.*, etc.

24.8 Ecological significance of *Trichoderma*

Trichoderma plays an immense role in creating a pollution-free environment through multivarious action. Based on the well-established enzyme profile in *Trichoderma*, it is used for

- Plant growth promotion and disease control
- Improve soil health
- Bioremediation of polluted soils
- Industrial applications
- Phytobial remediation

24.8.1 *Trichoderma* and plant health

Trichoderma species have been widely exploited for the management of seed and soil-borne diseases of agricultural and horticultural crops (Tables 24.1 and 24.2). Soilborne diseases due to *Rhizoctonia solani*, *Macrophomina phaseolina*, and *Fusarium* spp. causes a high percentage of mortality of the plants and consequent yield losses in

Table 24.1 Effect of *Trichoderma* spp. against plant diseases (Kannaiyan et al., 2001).

<i>Trichoderma</i>	Pathogen/disease	Crop	Delivery system
<i>Trichoderma</i> sp. C62	<i>Sclerotium cepivorum</i>	Onion	Soil application as bran sand inoculums
<i>Trichoderma</i> spp.	<i>G. graminis</i> var. <i>tritici</i>	Wheat	Spores added to soil
	<i>R. solani</i>	Lettuce	Soil
<i>T. harzianum</i>	<i>Fusarium graminearum</i>	Lettuce	Soil
		Radish	Soil
<i>T. harzianum</i> T951	<i>Pythium ultimum</i>	Cucumber	Soil
<i>T.koningii</i>	<i>S. rolfii</i>	Tomato	Soil
<i>T.koningii</i>	<i>Pythium</i> spp.	Chickpea	Seed
	<i>Pythium</i> spp.	Pea	Seed
	<i>P. ultimum</i>	Pea	Soil
<i>T. harzianum</i>	<i>Pythium ultimum</i>	Cotton	Seed
		Maize	Seed
		Pea	Seed

Table 24.1 Continued

<i>Trichoderma</i>	Pathogen/disease	Crop	Delivery system
<i>T. harzianum</i> +Rhizobium	<i>S. rolfsii</i>	Groundnut	Seed and soil
<i>T. viride</i>	<i>M. phaseolina</i>	Sesamum	Seed
<i>T. harzianum</i>			
<i>T. longibrachiatum</i>			
<i>T. viride</i>	<i>M. phaseolina</i>	Groundnut Sesamum Blackgram Greengram Pigeonpea Sunflower	Seed
<i>T. hamatum</i>	<i>R. solani</i>	Cotton Sugarbeet Radish	Soil-as mycelial preparations
<i>T. viride</i>	<i>M. phaseolina</i>	Blackgram	Seed
<i>T. viride</i>	<i>M. phaseolina</i>	Greengram	Furrow application
<i>T. viride</i>	<i>F. udum</i>	Pigeon- pea	Seed and Soil
<i>T. viride</i>	<i>M. phaseolina</i>	Banana	<i>Trichoderma</i> capsules
	<i>F. oxysporum</i> f. sp. <i>cubense</i>		Seed
	<i>R. solani</i>	Cotton	Seed
	<i>M. phaseolina</i>	Green gram	Seed +Soil
	<i>Pythium ultimum</i>	Cowpea Chilies, Tomato	Solid matrix priming
<i>Trichoderma</i> spp.	<i>M. phaseolina</i>	Sesamum Blackgram	Seed Seed and soil
<i>T. viride</i>	<i>R. solani</i> <i>S. rolfsii</i>	Cotton Brinjal Sunflower Bhendi	Seed

rain-fed crops especially in pulses and oilseeds. Soil incorporation of *T. harzianum* in the field reduced the inoculum density of *R. solani* by 85% and the fruit rot of tomato by 27%–51%. Soil application of *T. harzianum* was found effective in reducing the incidence of *Macrophomina* root rot of beans by 37%–74%. Coating of melon seeds with conidia of *T. harzianum* reduced the disease caused by *Macrophomina* to the extent of 46.3%. Seed coating of green gram and cowpea at the rate of 10 g of *T. viride* isolate MG6 combined with soil application reduced dry root rot incidence and increased the yield (Nakkeeran and Doraisamy, 2001). In addition to disease reduction, dry matter production was also enhanced which was attributed to the production

Table 24.2 Commercial bio-products of *Trichoderma* spp. (Kannaiyan et al., 2001).

Product	Target pathogens	Crops recommended	Manufacturer
Bio-fungus- <i>Trichoderma</i> spp.	<i>Sclerotinia</i> , <i>Phytophthora</i> , <i>R. solani</i> , <i>Pythium</i> spp., <i>Fusarium</i> , <i>Verticillium</i>	Flowers, Strawberries, Trees, Vegetables	De Cuester, Belgium
Binab T– <i>T. harzianum</i> and <i>T. polysporum</i>	Pathogens causing wilt, take all, root rot, and internal decay of wood products	Flower, fruits, ornamentals, turf and vegetables	Bio-innovation Lab-UK
Root-Pro-T. <i>T. harzianum</i>	<i>R. solani</i> , <i>Pythium</i> spp., <i>Fusarium</i> spp., and <i>S. rolfsii</i>	Flowers and vegetables	Mycontrol Ltd., Israel
Root shield- <i>T. harzianum</i>	<i>R. solani</i> , <i>Pythium</i> spp., <i>Fusarium</i> spp.	Trees, Shrubs, cabbage, tomato, cucumber, and all ornamentals	Bioworks, Inc., USA
Supresivit- <i>T. harzianum</i>	Several fungal pathogens	Broad spectrum	Borregaard BioPlant, Denmark
T22G, T22 planter box- <i>T. harzianum</i>	<i>R. solani</i> , <i>Pythium</i> spp., <i>Fusarium</i> spp., and <i>Sclerotinia</i> <i>homeocarpa</i>	Bean, cabbage, corn, cotton, cucumber, peanut, potato, sorghum, soybean, sugarbeet, tomato, turf, and ornamentals	Bioworks, Inc., USA
Trieco-T. <i>viride</i>	<i>R. solani</i> , <i>Pythium</i> spp., <i>Fusarium</i> spp.	Cotton, chillies, tomato, sugarcane, citrus, grapes, sunflower, cereals, and vegetables	Ecosense labs (1) Pvt. ltd., Mumbai, India
Trichodex- <i>T. harzianum</i>	<i>B. cinerea</i> , <i>Colletotrichum</i> spp., <i>Fulvia fulva</i> , <i>Monilia laxa</i> , <i>Plasmopara viticola</i> , <i>Pseudoperonospora</i> <i>cubensis</i> , <i>Rhizopus stolonifer</i> , <i>S.</i> <i>sclerotiorum</i>	Cucumber, grape, Nectarine, Soybean Strawberry, sunflower, tomato	Makhteshim Chemical Works Ltd., New York

Table 24.2 Continued

Product	Target pathogens	Crops recommended	Manufacturer
Trichopel, Trichoject, Trichodowel, Trichoseal- <i>T. harzianum</i> and <i>T. viride</i>	<i>Armillaria</i> , <i>Botryosphaeria</i> , <i>Chondrostereum</i> , <i>Fusarium</i> , <i>Nectria</i> , <i>Phytophthora</i> , <i>Pythium</i> , <i>Rhizoctonia</i>	Broad spectrum	Agrimm Technologies Ltd., Newzealand
<i>Trichoderma</i> 2000- <i>Trichoderma</i> spp.	<i>R. solani</i> , <i>Pythium</i> spp., <i>Fusarium</i> spp., and <i>S. rolfsii</i>	Nursery and field crops	Mycontrol Ltd., Israel

of growth hormones. Treatment of sunflower seeds with *T. viride* at the rate of 4 g/kg reduced the root rot incidence to 18.15 as against 37.1% in control. It increased the yield of 21%.

Seed treatment with *T. viride* @ 4 g/kg reduced the root rot incidence of sesame to 12.8% and increased the root, shoot length, and oil content. It recorded a yield of 677 kg/ha as against 301 kg/ha in control. Results of the field demonstrations conducted in the farmer's fields for the control of root rot in groundnut, sesamum, urdbean, and chickpea against dry root rot through *T. viride* was found to be promising. *T. harzianum* was found effective in the management of *Fusarium* wilt diseases. Strain T35 checked the *Fusarium* wilt of cotton and melon under natural soil conditions. Incorporation of *T. harzianum* as wheat bran-peat preparation into the soil under greenhouse conditions reduced the incidence of crown rot of tomato up to 80%. Soil application increased the yield of tomato by 18.8%.

Soil application of *T. koningii* at the rate of 2 g per vine of black pepper at the time of transplanting was highly effective than metalaxyl compounds against *Phytophthora* foot rot. The disease incidence was 23%–35% lower and yield was 30%–40% higher than control. Pre- and postplanting application of *T. viride* and *T. harzianum* was effective against foot rot disease of the beetle vine. Amendment of nursery soils with *T. harzianum* prior to the colonization of damping off pathogen in nursery beds resulted in the reduction of pre and postemergence damping-off of cardamom. Delivery of *Trichoderma* strains through solid matrix priming in tomato and chillies reduced the incidence of damping-off under field conditions than dry seed treatments (Nakkeeran and Doraisamy, 2001; Kannaiyan et al., 2001).

24.9 Factors influencing bioefficacy of *Trichoderma* in maintaining plant health

The soil physicochemical factors like pH, moisture, temperature, soil type, components of soil atmosphere, inorganic and organic constituents of soil, and the quantity and type of pesticides applied to soil largely influence the bio-efficacy of *Trichoderma*.

24.9.1 pH

Hydrogen ion concentration played a major role in the adaptability of biocontrol agents to their nutrient environment. The activities of *Trichoderma* spp. were more pronounced at low pH levels and the efficacy was influenced by acidic conditions (Chet and Baker, 1980). The activity of *T. lignorum* was poor in neutral and alkaline soils. Acidic pH favored disease suppression induced by *T. harzianum* (Chet and Baker, 1980) and *T. hamatum*. The spore germination was enhanced under acidic conditions (Chet and Baker, 1980), mycelial growth, conidiophore production (Chet and Baker, 1980), antibiotics production (Dennis and Webster (1971), or the activity of lytic enzymes (Chet and Baker, 1980). Jeyarajan et al. (1994) found that the isolates of *T. viride* performed on a wide range of pH varying from 5.0 to 9.0. Since the isolates of *T. viride* perform in a wide range of soil pH, a composite, compatible formulation of *T. viride* could be developed and used in disease management programs.

24.9.2 Moisture

The soil moisture content is very important compared with soil reactions in altering the population dynamics of the soil. It affects the growth of the antagonist. The bacterial population dropped rapidly as the soil moisture was falling below 15%–20% while fungi and actinomycetes population were not affected. The minimum relative humidity required for the growth of *Trichoderma* was 91%. The maximum growth of *Trichoderma* on ground pine needles occurred at 144% and 201% moisture. Growth was good at 103% moisture but was depressed at 46% moisture. Growth of *T. viride* was more at 40%–60% moisture than at higher soil moisture levels. Jeyarajan et al. (1994) observed that all the isolates of *T. viride* and *T. harzianum* survived uniformly well at 40% followed by 60% moisture-holding capacity.

24.10 Mode of action

Antagonist interacts with pathogen and host in soils either directly or indirectly. In indirect interactions, plant responds to the presence of the antagonist, resulting in induced resistance or plant growth promotion. Based on the interaction, the modes of action may be direct or indirect.

24.10.1 Direct mode of action

24.10.1.1 Competition

Most of the soilborne pathogens are controlled by the competition for space or court of infection on roots and seeds. Biological control of silver leaf disease caused by *Chondrostereum purpureum* was controlled by *T. viride* through the early colonization of fresh wounds (Corke and Hunter, 1979). Competition also seems to be the most potent mechanism of *T. harzianum* T-35 in the control of *F. oxysporum* f. sp. *vasinfectum* and *F. oxysporum* f. sp. *melonis* in cotton and melon, respectively (Sivan and Chet, 1989).

24.10.1.2 Antibiosis

Antibiotics are low-molecular-weight (<1 kDa) secondary metabolites produced during nutrient limiting conditions. Gliotoxin produced by *G. virens* G20 is a key factor in reducing damping-off of *Zinnia* (Wilhitte et al., 1994). *G. virens* pertaining to 'Q' group produced gliotoxin and dimethyl gliotoxin, but 'P' group produced gliovirin (diketopiperazine) and heptelidic acid. Gliovirin was inhibitory to *P. ultimum* which causes damping-off in cotton compared with 'Q' strains, which was effective against *R. solani* (Howell et al., 1993).

Trichoderma produced several kinds of antibiotics like Trichodermain, dermadin, trichoviridin, and sesquiterpene heptalic acid. The UV mutants of *T. harzianum* produced trichozianines, the hydrophobic peptides, which interact with phospholipid membrane thus inducing membrane permeability. *T. hamatum* produced trichoviridin 3-(3-isocyano-6-oxabicyclo (3,10) hex-2-ch-5yl) and acrylic acid 3-(3-isocyano cyclopet-2-enzyldidne). In addition, it produced three isonitriles among which isonitrin A had broad-spectrum activity. *T. longibrachiatum* produced Trichodermain and trichobrachin (Jeyarajan and Nakkeeran, 2000).

24.10.1.3 Hydrolytic enzymes

Cellulases

Trichoderma species were potential biocontrol agents and producers of powerful cellulases (Papavizas, 1985). Mandels et al. (1971) irradiated *T. viride* and produced a new strain that secreted twice as much cellulase as its wild parent. Improved strains of *T. reesei* were isolated through mutation and selection. It produced more cellulase than did wild strains (Montenecourt and Eveleigh, 1979). Baker (1991) found a perfect correlation between increased cellulase production and rhizosphere competence by mutants of *T. harzianum*. It is suggested that mutants use the mucigel on roots as a substrate. Mutants with rhizosphere competence were a more efficient biocontrol agent and induced increased growth responses from seed treatment than wild-type parents. The quality and quantity of cellulases from *Trichoderma* spp. were improved through mutagenesis (Cuevas et al., 1994).

Proteases

Trichoderma species are well-known producers of proteases. Proteolytic activity of *T. viride* was claimed to be involved in biocontrol of *Sclerotium rolfsii* (Rodriguez-Kabana et al., 1978). Serine proteases produced by *T. harzianum* played an important role in mycoparasitism of plant pathogens (Geremia et al., 1993). Autoclaved mycelium, fungal cell wall, or chitin induced this enzyme, and its production was repressed by glucose. Proteases are suggested to degrade cell wall, membranes, and proteins released by the lysis of the pathogen, thus making the nutrients available for the mycoparasite (Goldman et al., 1994).

24.10.1.4 α -glucosidases

It is an *exo*-carbohydrase that catalyzes the splitting of α -glucosyl residues from the nonreducing terminus of substrates to liberate α -glucose. α -glucosidase is a key enzyme used in the food industries. But its role in the inactivation of phytotoxins

has been recently exploited for disease management. *T. viride* produces 67 kDa protein with a pI value of 5.0. *Rhizoctonia solani* causing rice sheath blight produced a glycoproteinaceous host-specific RS toxin. α -glucosidase of high molecular weight (110 kDa) from *T. viride* isolate (Tv-MNT7) was reported to degrade the RS toxin (Sriram et al., 2000).

24.10.1.5 β -1,3 glucanases

Lytic enzymes play a vital role in mycoparasitism and degrade the pathogen cell wall. The enzymes like chitinases and β -1,3 glucanases lyse the host wall and lead to the leakage of protoplasmic contents, which are in turn used as food material for the multiplication of antagonist (Cherif and Benhamou, 1990; Elad et al., 1983). Strains of *T. harzianum* produced a mixture of lytic enzymes comprising of chitinolytic and gluconolytic enzymes, which involved in mycoparasitism (Sivan and Chet, 1989). The excretion of β -1,3 glucanase into growth medium and to the soil by *T. harzianum* was observed (Elad et al., 1982). Carbon sources like laminarin, pustulan, cell walls of *R. solani*, and mycelia of several fungi incorporated in the enzyme production medium induced the secretion of β -1,3 glucanase by *T. harzianum* (Jacobs et al., 1991). The purified endochitinase, chitobiosidase, *N*-acetyl(-glucosaminidase and glucan, 1,3- β -glucosidase, and combinations inhibited the spore germination and germ tube elongation of *Botrytis cinerea*. Combinations of the purified enzymes resulted in the synergistic increase in antifungal activity (Lorito et al., 1994a,b). High synergistic activity was observed by combining 1,3- β -glucosidases and chitinolytic enzymes. This corresponds to the effect observed with the combination of β -1,3 glucanase and endochitinase isolated from plants (Mauch et al., 1988). *T. harzianum* also produced *N*-acetylglucosaminidase, endochitinase, chitobiosidase, and *endo*- β -1,3 glucanases in soybean rhizosphere active against *R. solani* (Dalsoglio et al., 1998).

Chitinases

Chitinase, the basic protein catalyzes the hydrolysis of chitin. Chitin a linear polymer of β -1,4-*N*-acetyl glucosamine is often considered to be the second most abundant polysaccharide in nature (Deshpande, 1986; Nicol, 1991). Relationship between mycolytic enzymes, chitinases, and β -1,3 glucanases produced by mycoparasitic fungi and their significance in fungal cell wall lysis and degradation have been well established (Elad et al., 1980, 1983). The assay of chitinase could be used as a basis for screening potential biocontrol agents (Elad et al., 1982). *Trichoderma* spp. attack several plant pathogenic fungi by excreting lytic enzymes including β -1, 3 glucanase, proteinases, and chitinases enabling them to degrade host cell walls and thus reduce disease incidence (Geremia et al., 1993; Chet et al., 1993; Lorito et al., 1993; Goldman et al., 1994). *T. harzianum* strain Tm had six distinct chitinolytic enzymes. Two were identified as β -1, 4-*N*-acetyl glucosaminidases (Ec. 3.2.1.30). It hydrolyzes chitin to *N*-acetyl glucosamine (Glc NAc) monomers. Four were endochitinases (Ec 3.2.1.14) that randomly cleave at internal sites over the entire length of the chitin microfibril (CHIT 52, CHIT 42, CHIT 33, and CHIT31). *T. harzianum*—P1 released chimeric units from chitin and termed as chitobiosidase (40 kDa) (Harman et al., 1993).

The involvement of *Trichoderma* chitinases in mycoparasitism was tested in liquid cultures supplemented with different carbon sources, e.g., chitin, purified fungal cell walls, glucose, or GlcNAc (Garcia et al., 1994). Complete degradation of mycelial walls of pathogenic fungi depends to major extent on chitinolysis and hence the degrading ability to chitin by *Trichoderma* spp. assumes a greater significance in antagonistic process than the other mechanisms/enzymes involved in the process (Cherif and Benhamou, 1990). Chitinases and glucanases were produced by *Trichoderma* isolates, in soil, which had live mycelium of *S. rolfssii* and *R. solani*. The high level of β -1,3 glucanase and chitinase activities were also detected by dual agar cultures of *T. harzianum* with *S. rolfssii* and *R. solani* (Elad et al., 1983). Direct interaction of *T. harzianum* with *R. solani*, strongly enhanced the endochitinase CHIT 42 (Carsolio et al., 1994). The spore germination and hyphal elongation of *B. cinerea*, *F. solani*, *F. graminearum*, *Ustilago avenae*, and *Uncinula necator* were inhibited substantially by the combination of chitinolytic and gluconolytic enzymes purified from *T. harzianum*. The chitinolytic enzymes from *T. harzianum* were more active than that of the enzyme from other sources like plant-bacteria and were effective against a wider range of fungi (Harman et al., 1993).

24.10.2 Spatial distribution of glucanases and chitinases

β -1,3 glucanase degrades the substrate β -1,3 glucan, which is the major component of the fungal cell wall next to chitin. The physiological concentration of these fungi effectively inhibits the growth of many potential pathogenic fungi. Though this enzyme plays a major role in cell wall lysis, it is imperative to understand the in vitro production and subcellular localization of the constitutive expression of the enzyme, β -1,3-glucanase. Expression of β -1,3 glucanase by *T. harzianum* was observed in extracellular, periplasmic, cell wall-bound internal fractions during submerged fermentation in the absence of lactose and glucose as carbon source. In submerged culturing of *T. harzianum* with lactose, β -1,3 glucanase was found in all extracellular, cell wall bound, and in periplasmic fractions. The relative distribution of the enzyme in the cell is independent of the nature of carbon source and its concentration.

Chitinases are extracellular and intracellular. Chitinases are localized either outside or inside the cell and it is localized at subcellular level in various fractions including cell wall, periplasmic space, vacuoles, and membranes. Extracellular chitinases are more important than the intracellular chitinases, since the extracellular chitinases provide the first line of defense against plant pathogens.

24.10.3 Mycoparasitism

Mycoparasitism relies on the production of lytic enzymes by the mycoparasite for the degradation of cell walls of the host fungus. Isolates of *Trichoderma* spp. coiled around the hyphae of *R. solani*, which consequently lost their cell contents and collapsed (Chu and Wu, 1981). *Trichoderma* spp. attached to *S. rolfssii* or *R. solani* by hyphal coils, hooks, or appressoria. Lysed sites and penetration holes were found

in hyphae of the pathogen following the removal of parasitic hyphae (Elad et al., 1983). *Trichoderma* coiled over *Pythium* resulted in cell wall lysis, vacuolation, and coagulation of protoplasm (Lifshitz et al., 1986).

24.10.4 Indirect modes of action

24.10.4.1 Induced resistance

The induction of resistance in plants by *Trichoderma* spp. has been poorly studied. The first demonstration of induced resistance by *Trichoderma* was published by Bigirimana (1997). Soil treatment with *T. harzianum* strain T-39 induced resistance against foliar pathogens *B. cinerea* and *Colletotrichum lindemuthianum*, even though T-39 was present only on the roots. *T. harzianum* strain T-22 induced systemic resistance to pathogens in maize. Therefore, the capacity to induce resistance to a range of diseases in a variety of plants seems to be widespread in *T. harzianum*.

24.10.4.2 Plant growth promotion

Growth promotion may occur either directly or indirectly. Promotion of growth may occur either through biocontrol of deleterious pathogens (Baker, 1989) or through the production of plant hormones like auxin, cytokinin, or gibberellins (Arshad and Frankenberger, 1991). Baker et al. (1984) found that *T. harzianum* can also stimulate plant growth, including floricultural and horticultural plants. Pepper seed germinated 2 days earlier in raw soils containing the fungus than in untreated controls. The flowering of periwinkle was accelerated, the number of blooms per plant on chrysanthemum was increased, and the height and weight of other plants were greater in either steamed or raw soil infested with *T. harzianum*. These responses have occurred consistently with population densities of *T. harzianum* higher than 10^8 cfu/g of soil, whether the biocontrol agent was applied as conidial suspension or as mycelium and spore in a peat bran medium.

Growth stimulation by *T. harzianum* could be the result of the production of plant hormones by antagonist, increased uptake of nutrients by plant, through nutrient mobilization, or the control of one or more subclinical pathogens. *Pythium* spp. are the capable of stunting plants and hinder the nutrient uptake of roots (Backer and Cook, 1988). Hence the suppression of *Pythium* spp. could increase plant growth (Salt, 1978). Increased growth was also induced by a diffusible growth-regulating factor produced by *Trichoderma* spp. (Windham et al., 1986).

24.11 Commercial production and formulations

It is the formulation that can transform a potential biological control agent from a laboratory curiosity to a commercially successful venture in terms of disease management. A successful biocontrol formulation is one, which is economical to produce, safe, stable in the environment, and easily delivered and should enable the biocontrol agent to act effectively and consistently under varied environmental conditions. The nutrient status of the substrate or additives is one of the important factors influencing

the efficacy of the formulation. Introduction of antagonists through organic carriers alleviate the competition from autochthonous microorganisms since the organic carriers serve as a protectant and a food base during the establishment of the antagonist. Delivering of *Trichoderma* spp. to the site of infection via the organic food bases like diatomaceous earth granule with molasses (Backman and Rodríguez-Kabana, 1975), wheat bran, wheat bran + sawdust (Elad et al., 1980), wheat bran + peat (Sivan et al., 1984), molasses yeast (Papavizas et al., 1984), alginate pellets (Fravel et al., 1985), pyrax (Papavizas and Lewis, 1989), vermiculite + wheat bran (Lewis et al., 1991; Nakkeeran et al., 1997), and processed manure pellets (Kok et al., 1996) allows the establishment and multiplication of the antagonist in the soil resulting in high population levels during an extended period of time.

24.11.1 Talc-based formulation

The Tamil Nadu Agricultural University, Coimbatore, has developed the technology of mass production of talc-based formulation of *Trichoderma viride* and *Pseudomonas fluorescens* for seed treatment and soil application (Sudisha et al., 2006). Several private industries in Coimbatore, Bangalore, Kerala, Maharashtra, Chethali, Delhi, and Chennai produce in large quantities. However, demand exceeds the present supply. The annual requirement of *Trichoderma* has been estimated as 5000 tons to cover 50% area in India. Indirectly, it also creates self-employment opportunities for the unemployed youths. The first commercial seed treatment formulation of the biocontrol agent in India was developed by Jeyarajan et al. (1994). *T. viride* was grown in molasses yeast medium for 10 days in flasks. Then, 2.5 L of the culture broth was inoculated to 50 L of sterilized molasses yeast medium in fermentor. It was incubated for 10 days with 4–8 h aeration/day. The fungal biomass and the broth were mixed with 100-kg talc (super white) powder and 500 g of carboxymethylcellulose as a sticker. It was dried in shade for 72 h and packed in alkathene bags. The initial population of *Trichoderma* in the product was 300×10^6 cfu/g. The product should contain a minimum of 20×10^6 cfu/g at the time of use. This is applied at the rate of 4 g per kg of seed. The shelf life was 4 months. Angappan (1992) found that in chickpea treated with this product, the rhizosphere population was maintained at $11\text{--}13 \times 10^3$ cfu per gram throughout the crop growth stage.

24.11.2 Gypsum

Ranganathan et al. (1995) found that gypsum was a good substitute for talc but was much cheaper than talc.

24.11.3 Industrial wastes

Nakkeeran and Jeyarajan (1996) tested two industrial wastes namely precipitated silica and calcium silicate in the place of talc and found them to give a population of 99, 104×10^6 cfu/g, respectively, compared to 143×10^6 cfu/g in talc substrate after 4 months of storage. These two substrates were also much cheaper than talc.

24.11.4 Diatomaceous earth granules

Diatomaceous earth granules were added into a broth consisting of 100-mL molasses, 900mL of water, 3 g each of KNO_3 and KH_2PO_4 till the level of saturation. It was autoclaved at 121°C for 15 min and spread in shallow pans to a height of 3–5 cm and autoclaved again. *T. harzianum* (3-day-old culture) was homogenized in a blender for 3 s and mixed with sterilized granules in shallow pans and incubated at 25°C for 4–7 days. Clumps were broken up and granules air dried with frequent stirring and used as an inoculum. The inoculum was mixed at 1:1 (v/v) with sterilized diatomaceous earth granules impregnated with 10% molasses solution. It was applied to peanut at 140 kg/ha on 70 and 100 days after sowing to control *Sclerotium rolfsii*. The disease was reduced by 42% over control and yield increased by 13.5% (Backman and Rodríguez-Kabana, 1975).

24.11.5 Wheat bran: Sawdust formulation

Wheat bran:Sawdust:tap water mixture (3:1:4 v/v) is taken in polypropylene bags and autoclaved for 1 h at 121°C for two successive days. The bags were inoculated with *T. harzianum* and incubated in illuminated chambers for 14 days at 30°C . It was applied at the time of sowing and mixed with the soil to a depth of 7–10 cm with a rotary hoe. It increased yield of beans (1500 kg/ha), tomato (300 kg/ha), cotton (500 kg/ha), and potato (40–600 kg/ha) and controlled *S. rolfsii* and *R. solani* (Elad et al., 1986).

24.11.6 Wheat bran: Peat

Wheat bran:peat mixture (1:1 v/v) was autoclaved for 1 h. Substrate moisture was adjusted to 50% (w/w) with sterile water, medium was inoculated with 0.1 mL of a conidial suspension containing 2×10^4 conidia/mL and incubated for 7 days at 30°C , and it was mixed with rooting mixture for a tomato at 10% v/v to control crown rot (*Fusarium oxysporum* f. sp. *radicis lycopersici*). The disease was reduced by 29.7% over control and yield increased by 7% (Sivan et al., 1984).

24.11.7 Vermiculite—Wheat bran

Trichoderma was multiplied in molasses—yeast medium for 10 days. Vermiculite (Grade 4) and milled wheat bran (250 mesh) were heated in a hot air oven at 70°C for 3 days using metal pans. Vermiculite (100 g), wheat bran (3.3 g), liquid culture (14 mL), and 0.05 N HCl (17.5 mL) were mixed and packed. This was immediately used for soil application (Lewis et al., 1991). Vidhya (1995) applied this formulation at 250 kg/ha to mung bean and found 41% reduction in root rot (*M. phaseolina*) and 91% increase in yield.

24.11.8 Alginate pellets

Sodium alginate (20 g) was dissolved in 750 mL water at 40°C on a stirring hot plate. Wheat bran ground to pass through a 0.425 mm mesh screen was placed in a glass blender container with distilled water (50 g/250 mL). Kaolin can also be used in place of bran. It was autoclaved for 30 min and cooled. Fermentor biomass (16–21 g/L) was added to provide 7×10^6 conidia and chlamydospores. The mixture containing fungus, alginate, and bran or kaolin was added dropwise into 500-mL gellant solution (0.25 M CaCl_2 , pH 5.4). As it entered, each droplet gelled, and a distinct spherical bead formed. After 20 min in the gellant, beads were separated from the solution by gentle filtration, washed, and dried for 24 h in a stream of air at 25°C (Lewis and Papavizas, 1986).

24.11.9 Other substrates

Farmyard manure, gobar gas slurry, press mud, paddy chaff, rice bran, and groundnut shell were also found to support the growth of *T. viride* and *T. harzianum* (Kousalya and Jeyarajan, 1988).

24.12 Shelf life

The shelf life of a biocontrol agent plays a crucial role in storage. In general, the antagonists multiplied in an organic food base has a greater shelf life than the inert or inorganic food bases. Viability of the encapsulated alginate pellets of *T. harzianum* isolate (Thr IDI) remained viable for at least 6 months when stored at 5°C and were reduced significantly when stored at 2°C (Dandurand and Knudsen, 1993). Jeyarajan et al. (1994) developed talc, peat, lignite, and kaolin-based formulation of *T. viride*, which had a shelf life of 4 months. The shelf life of the same in gypsum-based formulation was also 4 months but was cheaper than talc-based formulations (Ranganathan et al., 1995). Vermiculite bran acid fermentor biomass (VBA-FB) of *T. viride* recorded the highest mean population of 20.5×10^7 cfu/g up to 75 days of storage at room temperature (Nakkeeran et al., 1997).

24.13 Delivery system

Biocontrol agents can be applied to various plant parts like seed, root, foliage, inflorescence, and into the soil. Several effective methods of delivery have been developed which enhanced the yield of crops considerably (Nakkeeran et al., 1995; Nakkeeran and Renukadevi, 1997).

24.13.1 Seed treatment

Seed treatment is one of the methods for introducing biocontrol agents into soil–plant environment. The efficacy of seed treatment depends on the ability of the antagonist to multiply in the rhizosphere region. Seed treatment with *T. harzianum* reduced the disease caused by *R. solani* on cotton under field conditions (Elad et al., 1982). Treatment of black gram, green gram, pigeon pea, cowpea, groundnut, sunflower, sesamum, and cotton with *T. viride* at the rate of 4g/kg reduced the incidence of root rot and wilt (Dinakaran et al., 1995; Jeyarajan et al., 1994; Jeyarajan and Nakkeeran, 1996; Muthamilan and Jeyarajan, 1996; Nakkeeran and Renukadevi, 1997).

24.13.2 Solid matrix priming

It is a process in which moistened seeds are mixed with an organic carrier and the moisture content of the mixture is brought to a level just below that required for seed sprouting. Solid matrix priming can further enhance the efficacy of antagonists in treated seeds as it grows on seed surface during the priming process and thus the propagule numbers increase. The slurry of *Trichoderma* is first added to seeds followed by the addition of lignite and water. *Trichoderma* required less moisture for growth. The primed seeds were incubated for 4 days at 80%–100% RH. During this process, *Trichoderma* profusely sporulates on the seed surface. In *Pythium ultimum* infested soil, the stand was increased to 70%–80% as against 10% in control. In tomato seed, the population of *Trichoderma* increased by 10-fold due to solid matrix priming (Harman et al., 1989).

24.13.3 Soil

The high population of the antagonist in the soil is necessary for the effective management of soilborne pathogens. Adams (1990) defined the efficiency of biocontrol agents as the ratio of the number of propagules of mycoparasites required to obtain disease control to the typical inoculum density of a plant pathogen. Therefore, attempts were made to develop a suitable delivery system for soil. The addition of wheat bran-based inoculum to soil gave 80% reduction of root rot over control in chickpea and bean (Elad et al., 1986). The incidence of urdbean root rot was reduced by 91.3% by adding *T. viride* + *T. harzianum* to soil (Kousalya and Jeyarajan, 1988). Delivering *T. harzianum* through soil during sowing increased the survival of peanut (90%), while in control none of the plants survived (Muthamilan and Jeyarajan, 1996).

24.13.4 Cut stump application

Ricard (1981) developed a special shear to apply *T. viride* simultaneously with pruning to protect fruit trees against silver leaf disease caused by *Chondrostereum purpureum*. The shears have a reservoir for the suspension of *T. viride*. Each pruning cut was covered by spore suspension.

24.13.5 *Hive insert*

Thomson et al. (1992) developed an innovative method of application of biocontrol agents right in the infection court at the exact time of susceptibility. *Botrytis cinerea* (fruit rot of strawberry) was effectively controlled by *T. virens* through honeybees which acted as a vehicle to carry the spores of *Trichoderma*. Peng et al. (1992) developed a novel and more efficient method of applying *T. virens* on talc and cornmeal (5×10^8 cfug/g), which was placed inside the beehive in a dispenser. Bees acquired the conidia on their legs and bodies as they crawled. Each bee had a spore load of $88\text{--}1800 \times 10^3$. Each flower got 1.6 to 27×10^3 conidia of the antagonist. It suppressed *B. cinerea* on stamens (54%) and petals (47%) and controlled fruit rot.

24.14 Population dynamics of *Trichoderma*

24.14.1 *Soil*

The soil ecosystem is a hostile environment for introduced microorganisms, which often cannot establish and eventually disappear. The biocontrol efficacy of *Trichoderma* isolates seems to depend on the population size, reached after introduction into the soil. A successful antagonist, to survive better in soil and rhizosphere, should have the ability to proliferate well in the introduced environment (Baker and Cook, 1974). Population densities of *T. viride* (T-1-R4) and *T. harzianum* (WT-6-24) increased to 10^4 - and 10^3 -fold, respectively, in natural soil during the first 3 weeks of incubation (Lewis and Papavizas, 1984). The decline in the population of *T. viride* in the soil might be due to lysis and disintegration of conidia. The stable population densities of *T. viride* from 9 to 36 weeks of incubation may be due to the survival of chlamydospores. In no case, conidia of *Trichoderma* and *Gliocladium* did not proliferated in soil without the addition of a food base (Papavizas, 1985). Pyrax (Clay powder), alginate pellets, vermiculite-bran mixture-based formulations allowed establishment, and multiplication of the antagonist in soil, resulting in high population levels during the extended period of time. The activity and proliferation of *Trichoderma* spp. in the protection of radish seeds against *R. solani* was effective only in acidic soils (Chet and Baker, 1980). A food base such as ground wheat bran enables the fungal propagule in a given preparation to germinate and to colonize the soil. In sterile conditions, the propagation of *T. harzianum* was much greater than under unsterile soils. The incorporation of *T. harzianum* conidia through processed manure pellets helped colonization of nonsterile soils which indicated the necessity of food base for establishment. The soil pH also influenced population dynamics (Kok et al., 1996).

24.14.2 *Seeds*

Biocontrol agents must proliferate rapidly and become established in the place where they are applied. The first 12–24 h after planting is a crucial period for the bio-protectant applied as seed treatment against soilborne pathogens and they need to

be grown and established on the seed coat (Hubbard et al., 1983; Taylor and Harman, 1990; Nagaraju et al., 2012a,b). Solid matrix priming is the process, in which seeds are moistened and mixed with a finely ground organic carrier, and sufficient water was added to achieve the appropriate moisture potential for priming and then incubated for a given duration at a constant temperature. Organic carriers have been shown to enhance the efficacy of *Trichoderma* spp., the bio-agent proliferate during the incubation period and become established, more active, or have colonized the spermosphere thoroughly before planting (Harman et al., 1981). Primed seeds consistently emerge more rapidly than the untreated seeds and provide better stands under adverse environmental conditions than do untreated seeds. Solid matrix priming of tomato seeds increased the number of *Trichoderma* propagules by one order of magnitude in the spermosphere. The effectiveness of biocontrol agents may depend partially on their ability to proliferate during a short period of favorable environmental conditions before they encounter plant pathogens. The number of colony-forming units (cfu) of *Trichoderma* increased on seeds during the matrix priming in Agrolig by about 10-fold. The number of *Trichoderma* on matrix priming was 10^4 cfu per seed after 4 days of incubation, while on seeds treated with *Trichoderma* alone were about 10^3 cfu per seed. Hence the increase in the cfu of *Trichoderma* provided better emergence of tomato.

24.15 Strain improvement of *Trichoderma*

The existing strains can be further improved for achieving maximum disease control and for developing superior strains suited for industrial applications.

24.15.1 Genetic engineering

Genetic engineering is considered as one of the major scientific revolutions of the century. A successful biocontrol agent requires many characteristics. All of them may not be present in one strain. Since many biocontrol agents have infrequent/unknown sexual stages, crossing and progeny testing by conventional breeding is not possible. Genetic engineering is a useful tool for:

- i. Transfer, ability to produce antibiotics from one species to another.
- ii. Produce location-specific strains
- iii. Improve rhizosphere competence of the antagonist and
- iv. Improve its biocontrol potential

By these techniques, important attributes like tolerance to pesticides, adverse environmental conditions like extremes of moisture, temperature and pH, vigorous growth, sporulation in deep tank fermentation, and improved shelf life can be incorporated in one organism.

24.15.1.1 Methods of induction of new biotypes

New biotypes of biocontrol organisms can be developed by

- (i) Exposure to fungicides
- (ii) Exposure to chemical and physical mutagens
- (iii) Protoplast fusion and
- (iv) Protoplast transformation

24.15.1.2 Exposure to fungicides

T. harzianum when exposed to chlorothalonil, chlorothalonil-tolerant strains developed. They differed in morphological characters, growth habit, and sporulation from the wild type. Some showed enhanced biocontrol activity. This may be due to mutation, chromosomal aberration, or mitotic recombination. Prolonged and repeated exposure to fungicides like chlorothalonil, iprodione, procymidone, and vinclozolin has resulted in the selection of several strains of *T. harzianum* resistant to above fungicides.

24.15.1.3 Exposure to mutagens

Trichoderma has 109 alleles with great genetic variability and could be explored to induce improved strains from the existing wild ones. Evidence in the late 1970s showed that it was possible by physical and chemical mutagenesis to improve the yield and quality of enzymes by *Trichoderma* spp. (Montenecourt and Eveleigh, 1979). Improved strains of *T. reesei* were obtained by mutations and produced more cellulase than the wild parents (Mandels et al., 1971), but this technology was not imported to enhance the biocontrol potentialities of antagonistic fungi until 1980s.

24.15.1.4 Gamma irradiation

Mutants of *T. viride* resistant to benomyl were obtained through gamma irradiation, but its biocontrol properties were not reported (Papavizas and Lewis, 1983). Stable mutants of *G. virens* were obtained by exposing the conidia to gamma rays. It showed variations in phenotype, growth rate, sporulation, and antagonistic potential against *Rhizoctonia solani* and *Sclerotium rolfsii* from the parent strain. One mutant produced more chlamydospores but no conidia and was more aggressive than the wild type in antagonizing the pathogens in vitro. However, its performance under field conditions was not tested.

24.15.1.5 UV irradiation

The biotypes of *Trichoderma* differed considerably from wild type (WT-6) in appearance, growth habit, fungi toxic metabolite production against *Sclerotium cepivorum* and ability to survive in soil. Several genetic variants from wild-type strains of *T. harzianum* and *T. viride* were developed. Several biotypes tolerated high concentrations (500 µg/mL of the active ingredient) of benomyl and other fungicides like thia-bendazole and carbendazim. A few benomyl tolerant biotypes of *T. viride* and *T. harzianum* differed in their growth characters, ability to sporulate, and survival in soil and the suppression of saprophytic growth of *R. solani* in soil. Several

UV-induced biotypes were consistently more effective than the wild-type strains in quenching the activity of *P. ultimum* on peas, *R. solani* on cotton and radish, *S. rolfsii* on beans, and *S. cepivorum* on onion. The isolate (T-1-R9) was also effective against *Fusarium* wilt of chrysanthemum and *Rhizoctonia* scurf of potato (Beagle-Ristaino and Papavizas, 1985). Biotype T-1-R9 was able to recolonize the soil even in the presence of benomyl. It prevented the recolonization of pathogen and brought control similar to that of the integrated control that includes the use of benomyl, lime, and nitrate nitrogen. It was the most effective biocontrol agent against white rot of onion. The wild-type strain did not produce heat-labile metabolite to the assay organism (*S. cepivorum*).

24.15.1.6 Chemical mutagenesis

Exposing the conidia to *N*-methyl-*N*-nitro-*N*-nitrosoguanidine produced benomyl tolerant mutants of *T. harzianum* (T-95, T-12B). It produced more cellulase than the wild-type strain (Ahmad and Baker, 1987). They also found that the mutants utilized cellulose as a sole carbon source for growth and were able to colonize plant rhizosphere, whereas the wild-type strains did not. NTG induced mutants of *Trichoderma* spp. were tolerant to benomyl, but not rhizosphere competent (Ahmad and Baker, 1987).

Advantages

- (i) The development of such fungicide resistant/tolerant strains would encourage the industry to take up pilot testing and commercial production
- (ii) Could be combined with a minimal amount of fungicide for disease control.
- (iii) Such mutants can also be used in genetic recombination program by the introduction of the specific gene encoding for increased production of antibiotics and enzymes into a recipient biocontrol fungus that possesses other desirable traits.

24.15.1.7 Protoplast fusion

Protoplast fusion is an excellent approach to the construction of fungal strains with novel gene combinations. It is possible to combine desirable traits from various parental strains to produce superior biocontrol strains. Interspecific crosses have been accomplished in *T. viride*.

24.15.1.8 Protoplast transformation

It is the process by which naked DNA is introduced into a cell resulting in a heritable change. It involves two basic steps.

1. Partial or complete elimination of cell barriers to allow uptake of DNA into the cell
2. Stable integration, maintenance, and self-replication of DNA into cell. Isolated protoplasts are excellent materials for the introduction of foreign DNA. This can be done in three ways
 - i. Fusion of whole protoplasts from two strains, species, or genera.
 - ii. Transfer of mitochondrial plasmids-plasmid-mediated transformation. Hybrids contain genetic material only from the naturally inbreeding microbes from which crossed strains were derived.
 - iii. Transfer of DNA segments (autonomously replicating and integrated vectors)—recombinant DNA technique.

DNA segments are separated from the donor and inserted or attached to other cells. Eukaryotic genes cloned into *E. coli*, sequence is rearranged and introduced into the new organism. Resistance to benomyl was constructed by transformation (Ossanna and Mischke, 1990). This was accomplished by transferring to G1 21, a wild-type strain of *T. virens*, very sensitive to benomyl. The β -tubulin gene from *Neurospora crassa* that codes for resistance to benomyl was successfully expressed in *T. virens* strain GT-21. It suggests that there are tremendous possibilities for desirable transformations in biocontrol fungi provided that the genes coding for certain desirable traits have been identified.

24.16 Industrial application of *Trichoderma*

Trichoderma has been well explored for industrial applications such as bioremediation and enzyme production.

24.16.1 Bioremediation by *Trichoderma*

The variety of organic pollutants released by the ever-increasing number of industries is the direct cause of environmental and health-related problems that have detrimental effects on living beings. Conventional approaches (e.g., landfilling, recycling, pyrolysis, and incineration) to the remediation of contaminated sites are inefficient and costly and can also lead to the formation of toxic intermediates. Thus, biological decontamination methods are preferable to conventional approaches because, in general, microorganisms degrade numerous environmental pollutants without producing toxic intermediates. Some environmental pollutants, however, persist in the environment.

The many different organic pollutants that are present in the environment persist for varying periods of time. Depending on their half-life, some pollutants either disappear within a short period of time or transformed into nontoxic end products. Many others, however, such as polychlorinated dibenzodioxifurans (PCDD/Fs), dioxins, dichlorodiphenyltrichloroethane (DDT), chlordane, and hexachlorocyclohexane (HCH), have long half-lives in the environment and tend to enter the food web, where they are subsequently “biomagnified.” On entering mammalian systems, most of these pollutants are often metabolized to generate more toxic and harmful intermediates owing to the nonspecific activity of different mammalian enzymes. Bioremediation is the process by which living organisms degrade or transform hazardous organic contaminants. *Trichoderma* plays a vital role in the bioremediation of heavy metals, toxic chemicals such as xenobiotic compounds, cobalt, lignite, cyanide, nitrocellulose, etc.

Trichoderma is recognized as “keystone or controller organism” in forest soil due to their involvement in decomposition, nutrient cycling, and their regulation of associated mycobiota. Recent evidence have proved their potential to decompose xenobiotics. *T. harzianum* and *T. viride* are capable of degrading organochlorides under in vitro. The persistence of organochloride contaminants in forest soil may be influenced by the predominant association of *T. harzianum* and *T. viride*. *T. harzianum*

was capable of degrading chlorophenols and adsorbable organic halogens. It also could degrade ciliatine, glyphosate, and phosphonic acid (Lupicka et al., 1997).

24.16.2 Biodegradation of noxious chemicals

Trichoderma also has the potential to degrade noxious chemicals like nitrocellulose, cobalt 60, cyanides, and heavy metals. *Trichoderma reesei* could degrade nitrocellulose which is commonly used in explosives and pharmaceutical industries. The fungus was able to degrade nitrocellulose completely after 22 days of application (Auer et al., 2005).

24.16.3 Biodegradation of pesticide

T. harzianum degraded DDT, dieldrin, endosulfan, PCNB, and pentachlorophenol. *T. harzianum* CCT-4790 degraded 60% of the herbicide diuron in soil and opens a good potential for the bioremediation of herbicide-contaminated soils. *T. viride* degraded pesticide photodieldrin into water-soluble and noninsecticidal compounds after 4–5 weeks. Thus, *Trichoderma* strains play a crucial role in the bioremediation process. *Trichoderma harzianum*, a biocontrol species colonize plant rhizosphere and it metabolizes several pesticides. Pentachlorophenol (PCP) was metabolized to pentachloroanisole (Rigout and Matsumura, 2002).

24.16.4 Phytobiation

Physical and chemical techniques are the most commonly used methods for the remediation of pollutants. The high cost of these methods has stimulated the interest in alternative biologically based systems, in particular, bioremediation and phytoremediation. Bioremediation consists of adding microorganisms to a polluted system, in the presence of nutrients, for degrading toxic compounds. Phytoremediation uses plants to degrade or remove toxicants from soil lacking a defined or managed root-microbial population. The combined application of both remediations is termed as phytobiation. The cultivation of cotton, beans, or corn as well as ferns on a polluted site rich in heavy metals, arsenic, cyanide, and metallocyanides, nitrates, etc. in association with *Trichoderma* spp. increased rapid uptake of the toxic elements by the plants. In addition, there was a strong reduction of toxicity caused by polycyclic aromatic hydrocarbons and phenolic compounds in water and soil by *Trichoderma* (Vinale et al., 2003).

24.17 Conclusion

The investigation of the biogeographical distribution of fungi is a challenging task. Firstly, it is generally regarded that only a small proportion of fungi (approximately 70,000 species) have been identified to date, with the estimates of the numbers of fungi at 1.5 million. This estimate is not universally accepted, as it is based on

calculations from temperate regions indicating an approximately 6:1 ratio of fungal species to plant species. However, for tropical regions, there are conflicting estimates, with this ratio either considered too high or too low or proportions of up to 33:1 considered more realistic. Consequently, the incomplete knowledge of the diversity and distribution of fungi have to be taken into account in any fungal biogeography study. Owing to the significant contribution of *Trichoderma* to human welfare and due to the richness in the biodiversity of India, the following researchable issues have to be addressed to conserve and to claim the IPR rights of our microbial wealth. They are:

- Adoption of polyphasic systematics to map the distribution of *Trichoderma* from different agroclimatic zones
- Identifying the habitat preference of individual taxa
- Exploring the potential of all the strains of *Trichoderma* for human welfare
- Research on bioremediation, phytoremediation, and phytobiation has to be strengthened to select the potential strains of *Trichoderma*
- Transformation and gene manipulation strategies have to be strengthened for exploring the potential of *Trichoderma* other than *T. reesei*.
- Strengthening proteomic and genomic research of *Trichoderma* for better pest and disease control of plants.
- Improving the knowledge of the ecology of *Trichoderma*-plant and pathogen interaction.
- Assessment of human and plant health risks.
- Human Resource Development on bioinformatics, systems biology, digital biology with molecular techniques for the documentation, and exploitation of the full potential of *Trichoderma* for the welfare of the country and mankind.

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